

2024



Reducing Costs and Processing Times Using Amazon SageMaker with EagleView

Learn how EagleView in the software industry optimized the performance of its data extraction models using Amazon SageMaker.

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300–400% improvement

in model performance

40–50% reduction

in compute costs

90% reduction

in processing time

2-minute SLA

consistently met

99.999% uptime

achieved



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Overview

[EagleView](#) needed a solution to handle its increasingly complex machine learning (ML) needs. Amazon Web Services (AWS) for a scalable solution that could improve system performance and reduce costs.



EagleView migrated its data pipelines to [Amazon SageMaker](#)—where teams can build, train, and deploy ML models for any use case with fully managed infrastructure, tools, and workflows. As a result, it improved model performance by 300–400 percent, reduced compute costs by 40–50 percent, reduced processing time, met its customer service-level agreements (SLA) consistently, and improved overall system reliability.



Opportunity | Using Amazon SageMaker to Improve Scalability and Reliability for EagleView

EagleView uses aerial imagery combined with ML, computer vision, and analytics tools to provide insights to customers in the construction, real estate, insurance, emergency services, and energy industries. To collect images, EagleView operates a nationwide network of fixed-wing aircraft—covering 90–95 percent of the United States and a significant portion of Canada. These planes take images at ultra-high resolution four times more detailed than standard aerial photos and 70 times more detailed than satellite imagery. The fleet captures over 1 billion images and covers 9.5 million miles annually.

The company’s image-processing system then uses proprietary EagleView algorithms to identify and extract a wide variety of information. For example, it can detect structures and extract property attributes. It requires near real-time inferencing for certain business cases where there is a tight SLA that mandates less than a couple of minutes.

EagleView began as a small team in 2019, using only a few basic object detection ML models running on [Amazon Elastic Kubernetes Service](#) (Amazon EKS)—a managed Kubernetes service to run Kubernetes on AWS and on-premises data centers. The team’s Amazon EKS infrastructure was adequate to handle foundational ML models, but EagleView’s rapid growth and expansion into new industries necessitated more robust solutions to handle increasingly complex use cases.

As its ML demands grew, EagleView encountered challenges with large workloads where the system had to handle thousands of requests at once. The team began to have difficulties meeting its SLAs. So, it sought a performance ML solution that could help manage its large pipelines for image data extraction and use Amazon SageMaker for smooth integration with its other AWS services.

“We wanted to deploy our models in a standardized manner with automatic scaling,” says Bishwarup Bhattacharjee, head of ML at EagleView. “We needed a framework with many out-of-the-box capabilities, and that’s why we chose Amazon SageMaker.”

Solution | Achieving 300–400% Model Performance Improvement Using Amazon SageMaker

In 2021, the EagleView team started exploring using Amazon SageMaker and building a proof of concept. The company migrated its two pipelines over to Amazon SageMaker within 8 months. During the process, EagleView received support from the AWS team to quickly resolve some hosting issues. Additionally, EagleView participated in some training and discussion sessions with the AWS team to learn Amazon SageMaker features in depth.

“Our use cases were pretty complex at times, especially with the requirements of using different types of endpoints,” says Bhattacharjee. “The support that we received from AWS to navigate those challenges was amazing.”

The team had been using NVIDIA Triton Inference Server for deploying its computer vision models. Without having to navigate complex configurations, EagleView streamlined the migration of all its models by using the integrated NVIDIA Triton Inference Server containers on Amazon SageMaker.

Using its original infrastructure, the team needed to heavily configure and debug its models to handle large batches. Now, millions of images can be processed on EagleView’s large batch processing pipeline through Amazon SageMaker Inference. The team previously needed 16 hours to process 1,000 square miles of aerial images. After the migration, the team can process that volume in 1.5 hours, which represents a 90 percent reduction in processing time. This efficiency will help the team achieve its goal of processing 1 PB of data in 1 year.

“After migrating to Amazon SageMaker, we’ve increased the reliability of our processes,” says Bhavesh Savalia, solutions architect at EagleView. “Since we don’t have to spend time optimizing, we can operationalize our next pipeline and support larger workloads.”

By using near real-time inference endpoints on Amazon SageMaker, EagleView improved its model performance by 300–400 percent. The stability of the system also drastically increased, and the team could support inference loads without impacting SLA resolution. By using [Amazon SageMaker Asynchronous Inference](#), EagleView can queue incoming requests and processes them asynchronously—EagleView saved costs by autoscaling the instance count to



zero when there are no requests to process. Overall, EagleView has achieved 40–50 percent cost savings after the migration.

“We have saved significant costs and gained performance advantages by deploying our models using Amazon SageMaker,” says Savalia. “Now, we can scale according to our needs and deliver better performance using more cost-effective machines.”

Outcome | Creating a Holistic Environment for Future ML Applications

The successful migration to Amazon SageMaker is the first step of the journey for EagleView to realize the full potential of the AWS suite of ML solutions. The team plans to catalog their model artifacts in a model registry and begin using tools like Amazon SageMaker Studio for end-to-end ML development.

“Using Amazon SageMaker has opened the door to create a complete environment where everything is under one suite of products,” says Prem Kumar, CTO, insurance at EagleView. “We have developed a mature ML program to deliver high performance for all our pipelines.”

About EagleView

EagleView uses aerial imagery combined with machine learning, computer vision, and data analytics tools to provide insights to customers in construction, real estate, insurance, emergency services, energy, and many other fields.

AWS Services Used

Amazon SageMaker

Amazon SageMaker is built on Amazon’s two decades of experience developing real-world ML applications, including product recommendations, personalization, intelligent shopping, robotics, and voice-assisted devices.

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